

called *Packet Switching*. It's more efficient because it uses network resources only when data needs to be sent, and can even avoid crowded portions of the network to get data across faster.

Returning to GPRS: older GSM networks used the circuit switching approach to transfer data. This not only put a limit on the speed possible, it also didn't exploit the bandwidth on the network. GPRS uses packet switching - it throws data into the network, filling up any unused time-slots it finds. By exploiting the network thus, GPRS achieves speeds that weren't thought of in the old days of circuit switching.

As the number of calls increases, more and more TDMA channels get allocated to voice calls, leaving less free. This, alas, is where GPRS falters. It becomes slower as traffic in the cell increases. GPRS networks also do not allow for storing messages on the network. Unlike SMS, where the message can be stored and sent later if the network is busy, messages sent via GPRS are lost forever if they don't immediately reach the intended recipient.

GPRS, from the looks of it, looked to be the fastest way to transfer data on a GSM network, but it had one last step to take.

Enhanced Data Rates For GSM Evolution - EDGE

Put GPRS into high gear and you have EDGE. The big brother of GPRS, EDGE can be deployed over existing GPRS infrastructures. However, it requires better signal quality than what already exists on the world's GSM networks.

It uses a shiny new modulation technique to be able to pack in three times as much data into a packet as GPRS, achieving transfer rates of around 384 kbps for the common user - just enough to be called a 3G technology.

So there you have it. EDGE is the fastest and the last technology that will grace the GSM network. The future, all have realised, is CDMA.